

Stream API

1. Stock Data Analysis

Question:

You have a list of daily stock prices represented as objects with attributes like date, symbol, and closing price. Using the Stream API, how would you find the average closing price for a specific stock symbol over a given time period?

2. Transaction Filtering

Question:

Given a list of financial transactions, each containing information about the transaction type, amount, and date, how would you use the Stream API to filter and retrieve all transactions with amounts greater than a certain threshold?

3. Portfolio Diversification

Question:

You have a collection of investment assets with attributes such as risk level and return rate. How would you use the Stream API to filter assets that match a specific risk level and sort them by their return rate?

4. Currency Conversion

Question:

Assuming you have a list of currency exchange rates and a set of financial transactions in different currencies, how would you use the Stream API to convert all transactions to a specific target currency?

5. Transaction Aggregation

Question:

You need to summarize financial transactions based on different criteria, such as total amount spent per category. How would you use the Stream API to group transactions by category and calculate the total amount for each category?

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6. Historical Data Analysis

Question:

Given a dataset of historical financial data with attributes like date, price, and volume, how would you use the Stream API to find the date with the highest trading volume?

7. Risk Assessment

Question:

You have a list of loans with information about the loan amount, interest rate, and risk level. How would you use the Stream API to filter loans based on risk level and calculate the total loan amount for each risk category?

8. Credit Score Analysis

Question:

Assume you have a list of customers with their credit scores and loan amounts. How would you use the Stream API to filter and identify customers who meet specific credit score criteria for loan approval?

9. Investment Ranking

Question:

You are tasked with ranking a list of potential investments based on their return on investment (ROI). How would you use the Stream API to sort investments in descending order of ROI?

10. Monthly Expenditure

Question:

Given a list of monthly expenses with attributes like category and amount, how would you use the Stream API to calculate the total expenditure for each category over a specified time period?